

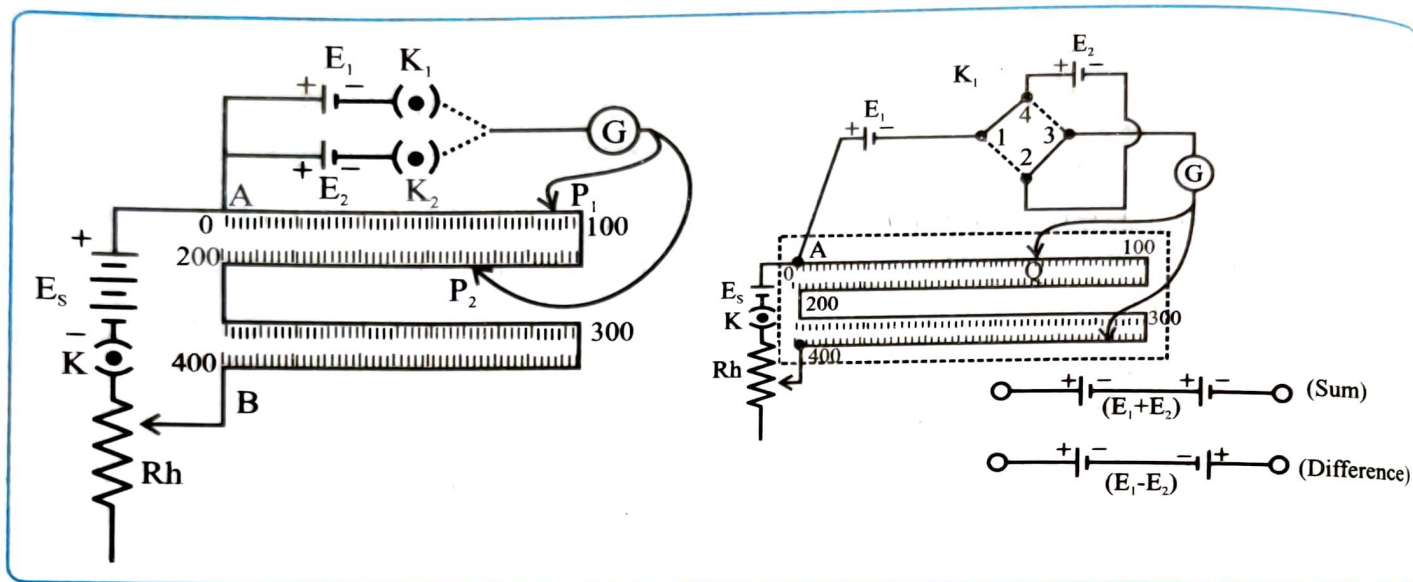
EXPERIMENT NO. 10

E₁/E₂ USING POTENTIOMETER

Aim: To compare e. m. f. of the two given cells using a potentiometer.

Apparatus: A potentiometer, a plug key, a rheostat, two cells, jockey, four way key, battery, galvanometer, connecting wires etc.

Diagram:



Formula:

1) Individual cell method or Separate cell method –

$$\frac{E_1}{E_2} = \frac{L_1}{L_2}$$

Where, L_1 is the balancing length when cell E_1 is in the circuit and L_2 is the balancing length when cell E_2 is in the circuit.

2) Sum and difference method –

$$\frac{E_1 + E_2}{E_1 - E_2} = \frac{L_1}{L_2}$$

Where, L_1 is the balancing length when both cells assist each other i.e. $E_1 + E_2$ (sum method) and L_2 is the balancing length when both cells oppose each other i.e. $E_1 - E_2$ (difference method).

Procedure:

Part – I (Individual cell Method or Separate cell method)

1. Measure the e.m.f. (E_s) of the battery and the e.m.fs. (E_1 and E_2) of the cells. See that $E_s > E_1$ and also $E_s > E_2$.
2. Connect the circuit as shown in fig.(i). Close the key K to bring the circuit in working, as current will start flowing. Adjust the rheostat at minimum resistance for maximum current in the circuit.
3. Take the cell E_1 in the circuit by closing key K_1 . Then touch the jockey at zero end of the potentiometer wire and note the direction of deflection in the galvanometer. Touch the jockey at the other end of the potentiometer wire. If the direction of deflection is opposite to that in the first case, the connections are correct. (If the deflection is in the same direction then either

connections are wrong or e.m.f. of the cell used is less). Starting from the point A touch the jockey on the wire at various points and obtain the null point P_1 . Note the length L_1 of the potentiometer wire between the point A and P_1 .

4. Now disconnect the cell E_1 by opening key K_1 . And connect the cell E_2 by closing key K_2 . Obtain the null point P_2 and note the length L_2 of the potentiometer wire between the points A & P_2 . Make sure that rheostat position is same as in step 2.
5. Repeat the observations alternately for each cell again by adjusting the rheostat at different positions.

Part -II (Sum and difference method)

1. Measure the e.m.f. (E_s) of the battery and the e.m.fs. (E_1 and E_2) of the cells. See that $E_s > E_1 + E_2$, $E_s > E_1 - E_2$ and $E_1 > E_2$.
2. Connect the circuit as shown in fig.(ii) such that the positive terminal of E_1 is connected to the negative terminal of E_2 . In this case both cells assist each other i.e. sum method. Close the key K to bring the circuit in working, as current will start flowing. Adjust the rheostat at minimum resistance for maximum current in the circuit.
3. Then touch the jockey at zero end of the potentiometer wire and note the direction of deflection in the galvanometer. Touch the jockey at the other end of the potentiometer wire. If the direction of deflection is opposite to that in the first case, the connections are correct. (If the deflection is in the same direction then either connections are wrong or e.m.f. of the cell used is less). Starting from the point A touch the jockey on the wire at various points and obtain the null point P_1 . Note the length L_1 of the potentiometer wire between the point A and P_1 .
4. Now connect cells, such that the negative terminal of E_1 is connected to the negative terminal of E_2 . In this case both cells oppose each other i.e. difference method. Obtain the null point P_2 and note the length L_2 of the potentiometer wire between the points A & P_2 . Make sure that rheostat position is same as in step 2.
5. Repeat the observations for each sum and difference method by adjusting the rheostat at different positions.

Observations :

E.M.F. of battery, $E_s =$ _____.

E.M.F. of cell, $E_1 =$ _____.

E.M.F. of cell, $E_2 =$ _____.

Part - I (Individual cell Method or Separate cell method)

Obs. No.	Balancing length when E_1 is in the circuit L_1 (cm)	Balancing length when E_2 is in the circuit L_2 (cm)	$\frac{E_1}{E_2} = \frac{L_1}{L_2}$
1.	228	196	1.163
2.	158	135	1.176
3.			
4.			
			Mean 1.1665

Part – II (Sum method and difference method)

Obs. No.	Balancing length when cells assist ($E_1 + E_2$) each other in the circuit L_1 (cm)	Balancing length when cells oppose ($E_1 - E_2$) each other in the circuit L_2 (cm)	$\frac{E_1 + E_2}{E_1 - E_2} = \frac{L_1}{L_2}$
1.			
2.			
3.			
4.			
			Mean

Calculations:

$$1) \frac{228}{196} = 1.163$$

$$2) \frac{158}{135} = 1.170$$

$$3) \frac{1.163 + 1.170}{2} = 1.1665$$

Result:

- 1) E_1/E_2 using separate or individual cell method =
- 2) E_1/E_2 using sum and difference method =

Precautions:

1. All the connections and plug keys should be tight.
2. In part – I, see that $E_s > E_1$ and $E_s > E_2$.
3. In part – II, see that $E_s > E_1 + E_2$, $E_s > E_1 - E_2$ and $E_1 > E_2$.
4. For a given set, the setting of the rheostat should be the same while determining the balance point P_1 and P_2 .
5. The balancing lengths should be measured from the point A.

Questions

1. What is an EMF of a cell?

The electromotive force (e.m.f.) of a cell is the potential difference between its terminals when no current is drawn from the cell.

2. What is potentiometer?

A potentiometer is an instrument used to compare the e.m.f. of two cells, measure the internal resistance of a cell, and measure potential difference accurately.

3. On what principle does the potentiometer work?

A potentiometer works on the principle that the potential drop across any length of a uniform wire carrying a constant current is directly proportional to its length.

Remark and sign of teacher :